

Module 8

Hints and Solutions

Section A – Fundamental Concepts

Problem 2

- (a) $|z| > 0.5$
- (b) $|z| < 2$
- (c) $0.5 < |z| < 2$, two sided sequence

Problem 3

- (a) $|z| > 1/3$, Exists
- (b) $|z| > 3$, Exists
- (c) $|z| > 0.5$, Exists
- (d) $|z| > 0.5$, Exists

Problem 4

- (a) ROC: Entire z -plane
- (b) ROC: $|z| > 1$
- (c) ROC: $|z| > 0.5$
- (d) ROC: $|z| > 1/3$

Problem 5

```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-5, 11)

impulse = np.where(n == 0, 1, 0)
step = np.where(n >= 0, 1, 0)
exponential = (0.5)**n * (n >= 0)
ramp = np.where(n >= 0, n, 0)

plt.figure(figsize=(10, 8))

plt.subplot(2,2,1)
```

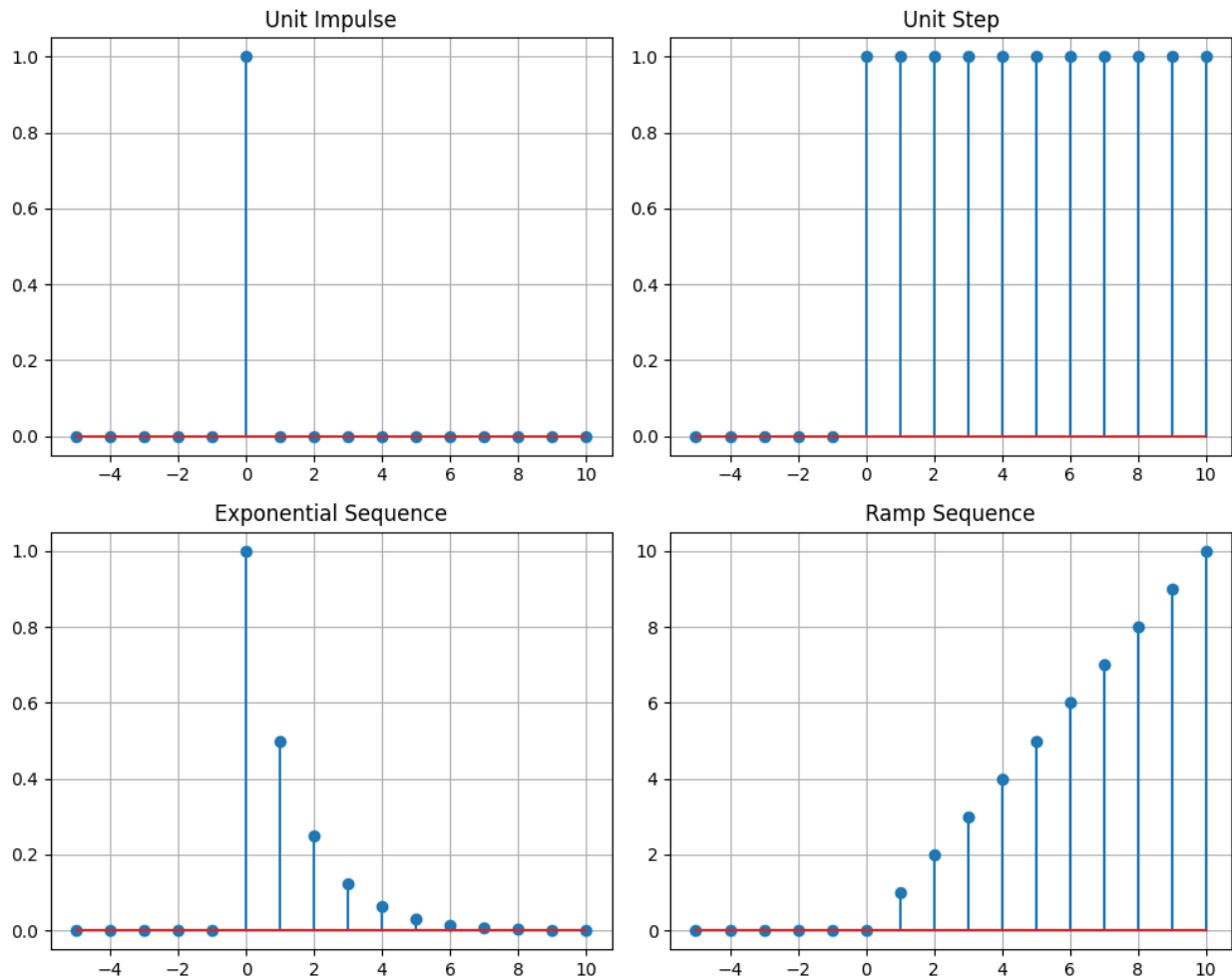
```
plt.stem(n, impulse)
plt.title("Unit Impulse")
plt.grid(True)

plt.subplot(2,2,2)
plt.stem(n, step)
plt.title("Unit Step")
plt.grid(True)

plt.subplot(2,2,3)
plt.stem(n, exponential)
plt.title("Exponential Sequence")
plt.grid(True)

plt.subplot(2,2,4)
plt.stem(n, ramp)
plt.title("Ramp Sequence")
plt.grid(True)

plt.tight_layout()
plt.show()
```



Section B – Properties of the Z-Transform

Problem 6

ROC: $|z| > 0.5$

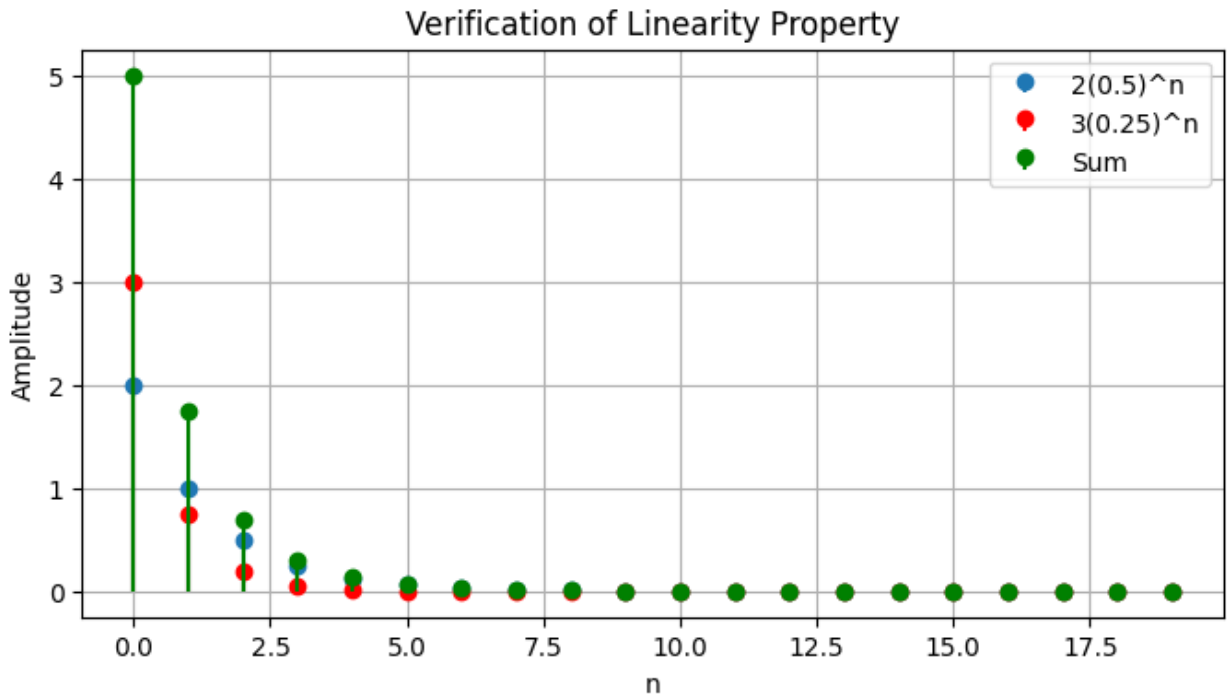
```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0,20)
x1 = 2*(0.5)**n
x2 = 3*(0.25)**n
x = x1 + x2
plt.figure(figsize=(8,4))

plt.stem(n,x1,label='2(0.5)^n',basefmt=" ")
plt.stem(n,x2,linewidth='r-',markerfmt='ro',label='3(0.25)^n',basefmt=" ")
plt.stem(n,x,linewidth='g-',markerfmt='go',label='Sum',basefmt=" ")

plt.xlabel("n")
```

```
plt.ylabel("Amplitude")
plt.title("Verification of Linearity Property")
plt.grid(True)
plt.legend()

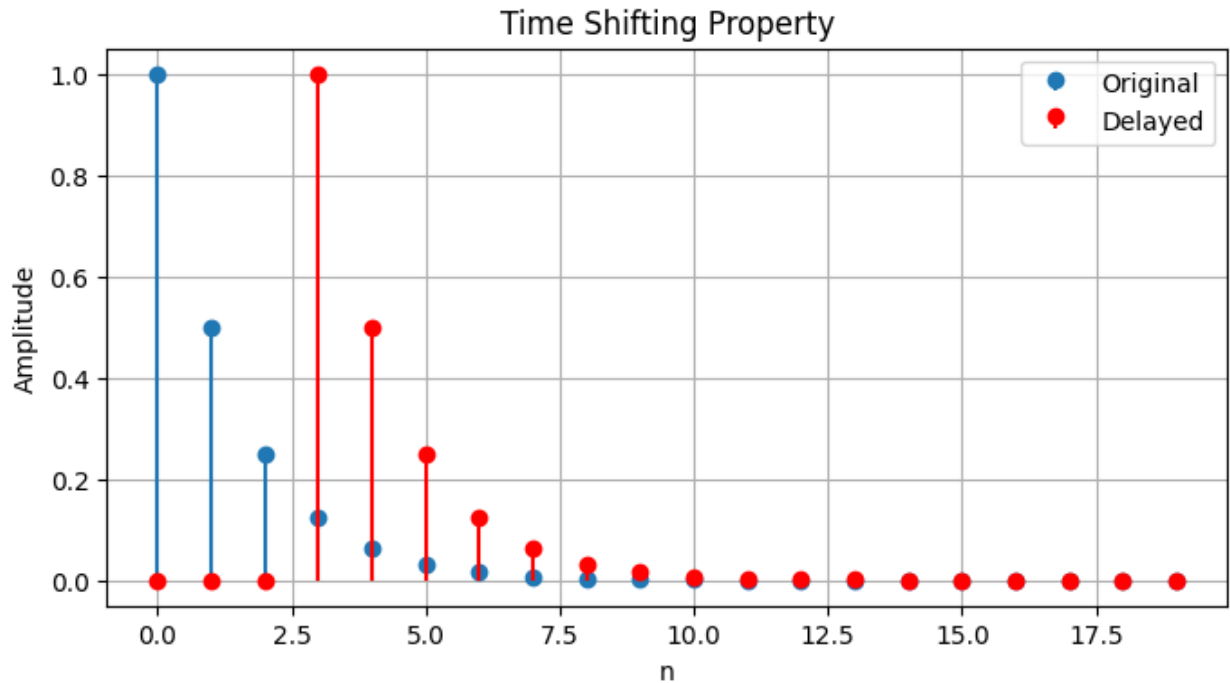
plt.show()
```



Problem 7

ROC: $|z| > 0.5$

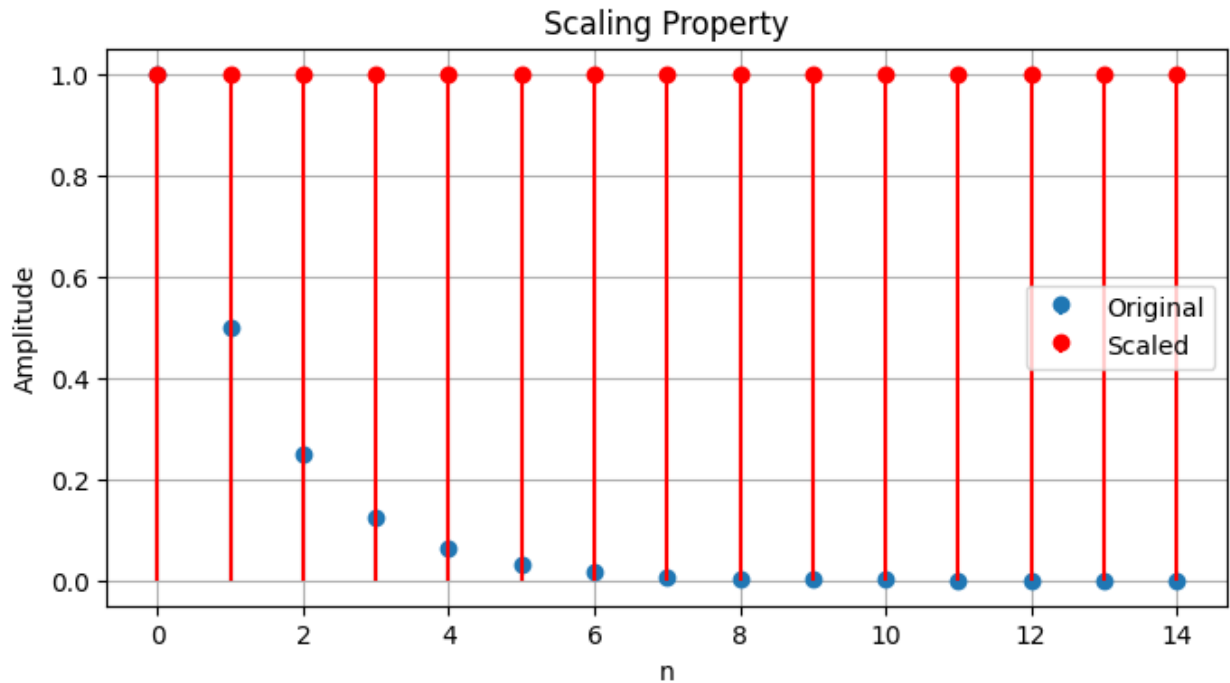
```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0,20)
original = (0.5)**n
delay = np.zeros_like(n,dtype=float)
delay[3:] = (0.5)**(n[:-3])
plt.figure(figsize=(8,4))
plt.stem(n,original,label="Original",basefmt=" ")
plt.stem(n,delay,linefmt='r-',markerfmt='ro',
        label="Delayed",basefmt=" ")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.title("Time Shifting Property")
plt.grid(True)
plt.legend()
plt.show()
```



Problem 8

ROC: $|z| > 1$

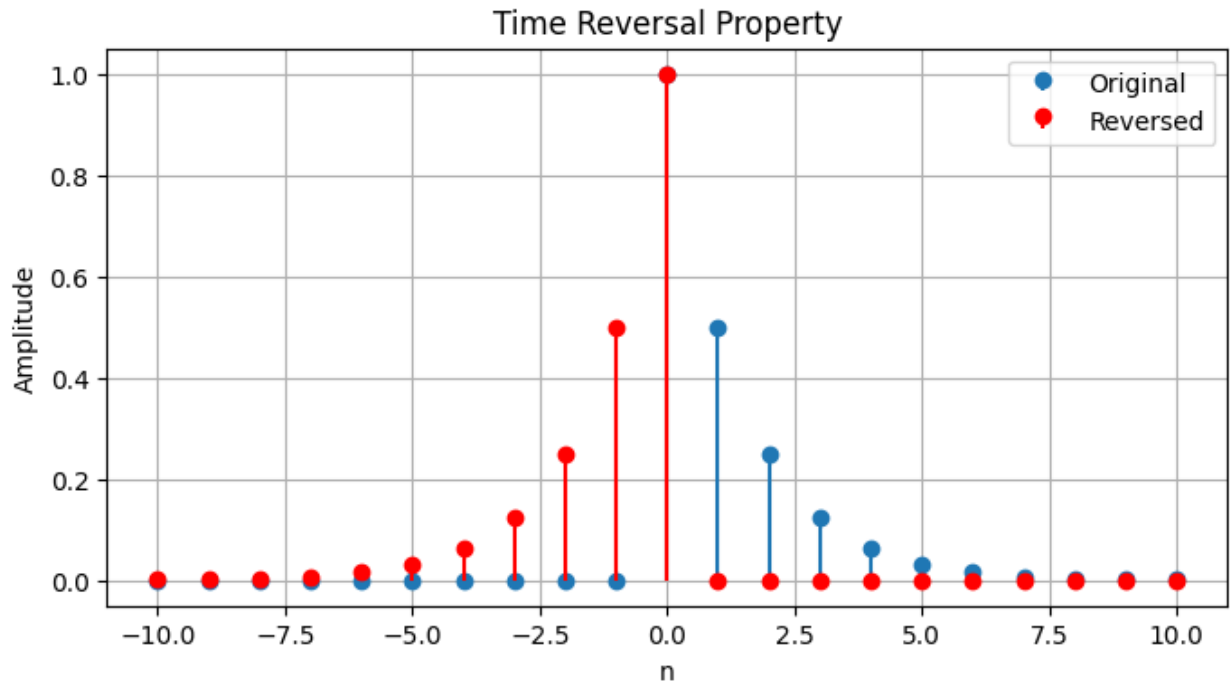
```
import numpy as np
import matplotlib.pyplot as plt
n=np.arange(0,15)
original=(0.5)**n
scaled=(2**n)*original
plt.figure(figsize=(8,4))
plt.stem(n,original,label="Original",basefmt=" ")
plt.stem(n,scaled,
         linefmt='r-',
         markerfmt='ro',
         label="Scaled",
         basefmt=" ")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.title("Scaling Property")
plt.grid(True)
plt.legend()
plt.show()
```



Problem 9

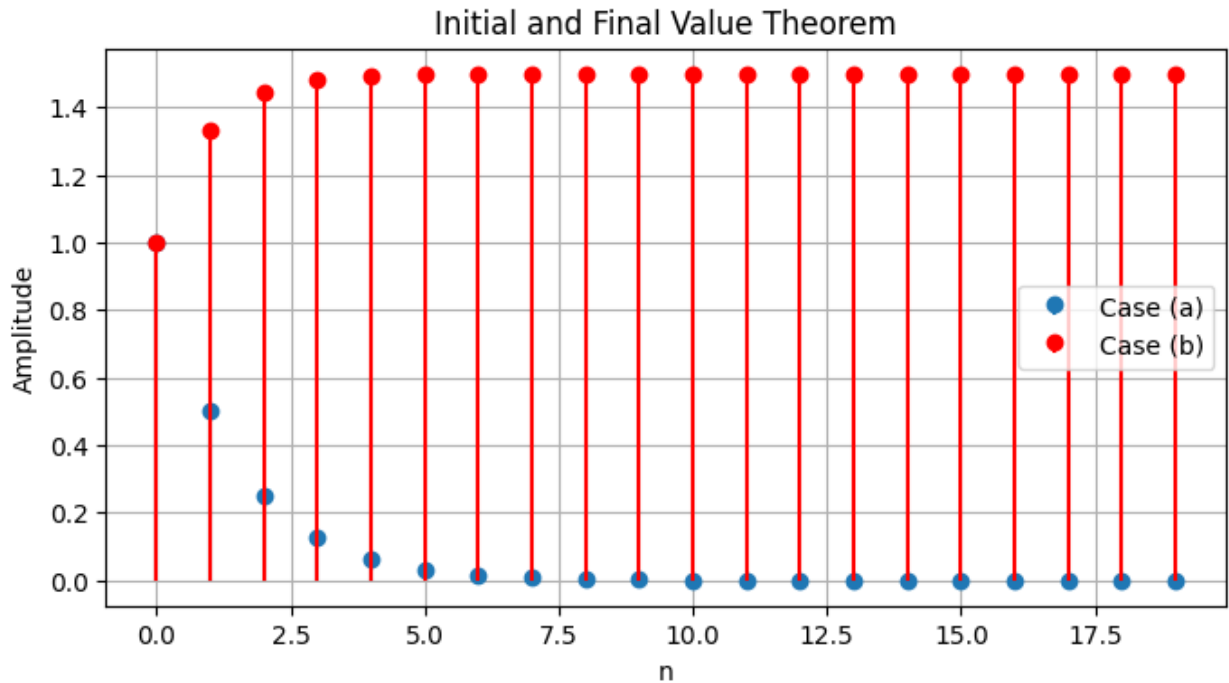
$$|z| < 2$$

```
import numpy as np
import matplotlib.pyplot as plt
n=np.arange(-10,11)
original=np.where(n>=0,(0.5)**n,0)
reverse=np.where(n<=0,(0.5)**(-n),0)
plt.figure(figsize=(8,4))
plt.stem(n,original,label="Original",basefmt=" ")
plt.stem(n,reverse,
         linefmt='r-',
         markerfmt='ro',
         label="Reversed",
         basefmt=" ")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.title("Time Reversal Property")
plt.grid(True)
plt.legend()
plt.show()
```



Problem 10

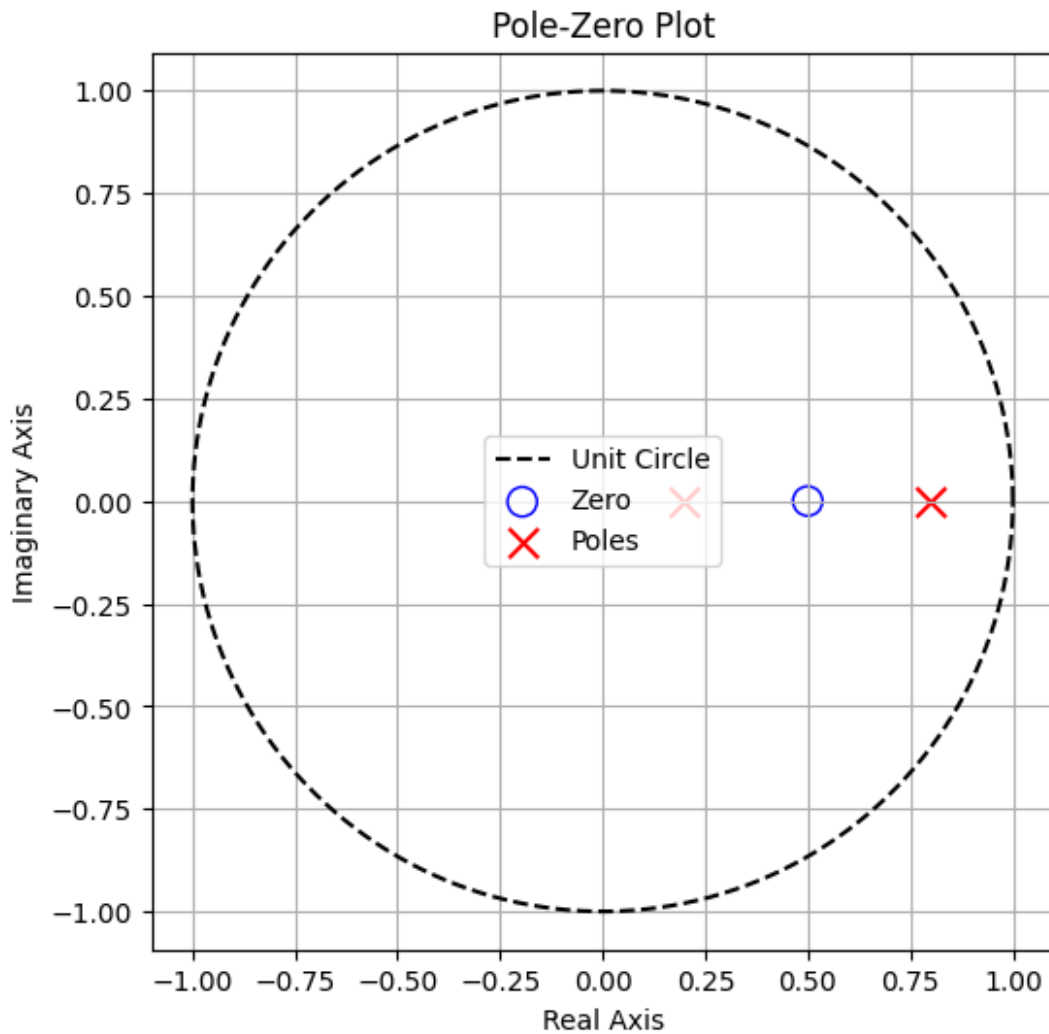
```
import numpy as np
import matplotlib.pyplot as plt
n=np.arange(0,20)
x1=(0.5)**n
x2=1.5*(1-(1/3)**(n+1))
plt.figure(figsize=(8,4))
plt.stem(n,x1,label="Case (a)",basefmt=" ")
plt.stem(n,x2,
         linefmt='r-',
         markerfmt='ro',
         label="Case (b)",
         basefmt=" ")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.title("Initial and Final Value Theorem")
plt.grid(True)
plt.legend()
plt.show()
```



Section C – Programming Problems

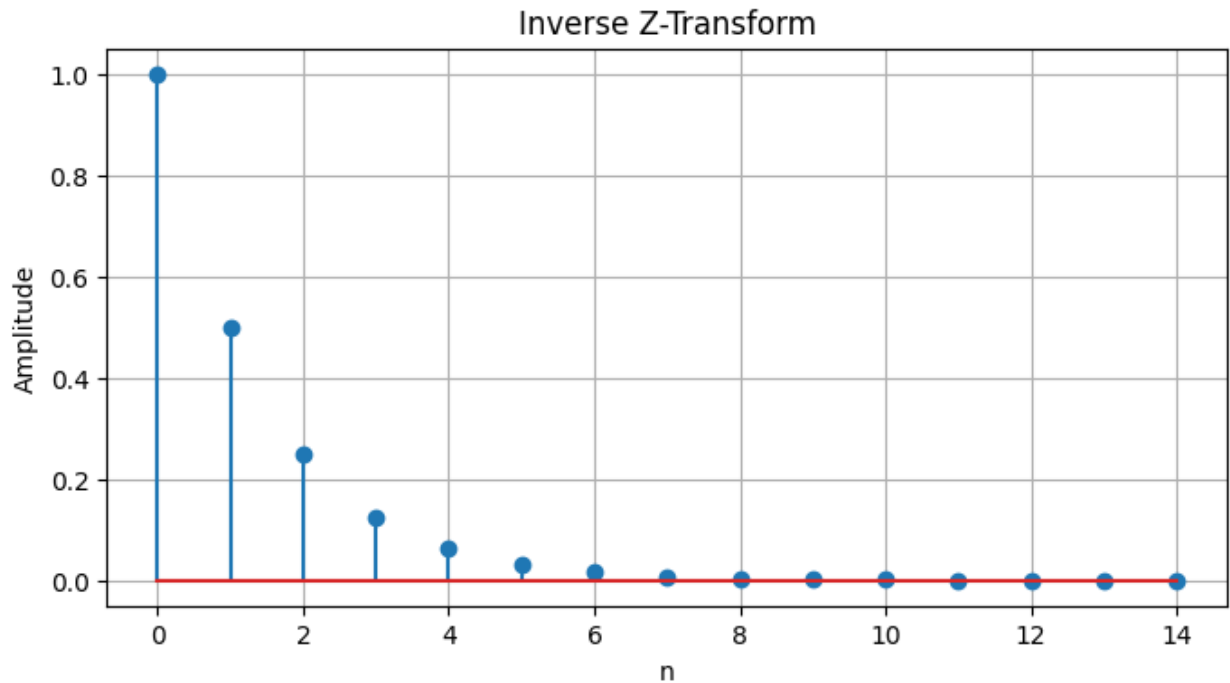
Problem 11

```
import matplotlib.pyplot as plt
import numpy as np
theta = np.linspace(0,2*np.pi,400)
plt.figure(figsize=(6,6))
plt.plot(np.cos(theta),np.sin(theta),'k--',label="Unit Circle")
plt.scatter([0.5],[0],
            marker='o',
            s=120,
            facecolors='none',
            edgecolors='blue',
            label="Zero")
plt.scatter([0.2,0.8],[0,0],
            marker='x',
            s=120,
            color='red',
            label="Poles")
plt.xlabel("Real Axis")
plt.ylabel("Imaginary Axis")
plt.title("Pole-Zero Plot")
plt.grid(True)
plt.axis('equal')
plt.legend()
plt.show()
```



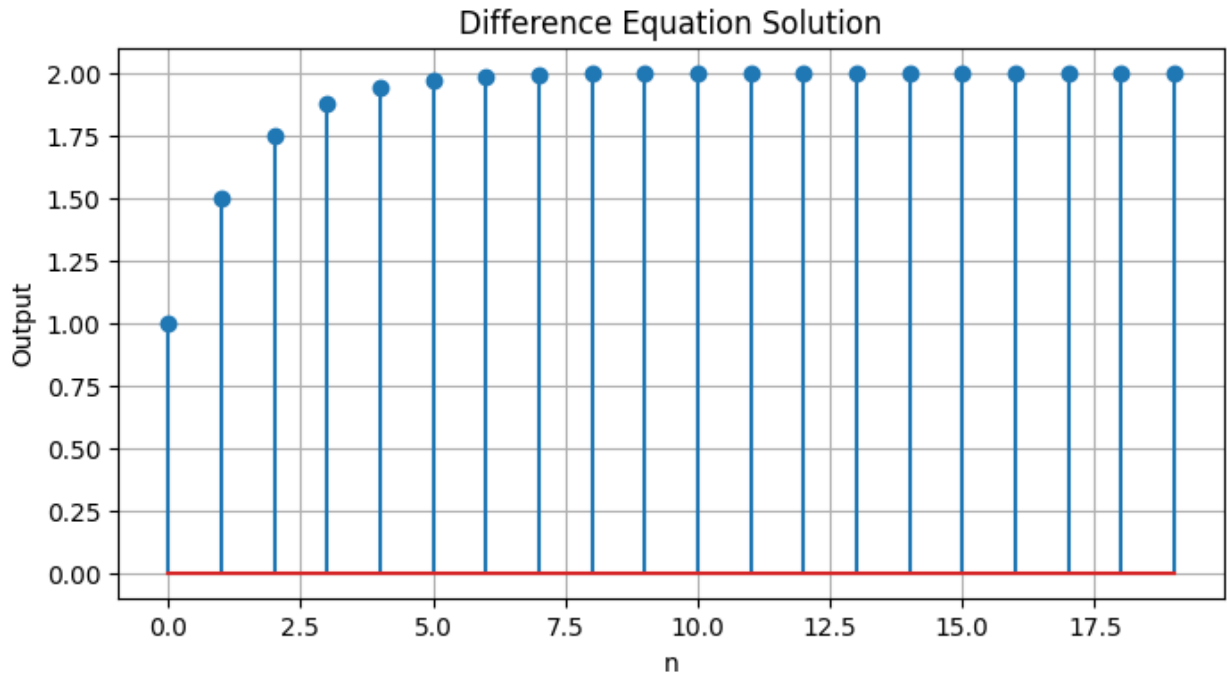
Problem 12

```
import numpy as np
import matplotlib.pyplot as plt
n=np.arange(0,15)
x=(0.5)**n
plt.figure(figsize=(8,4))
plt.stem(n,x)
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.title("Inverse Z-Transform")
plt.grid(True)
plt.show()
```



Problem 13

```
import numpy as np
import matplotlib.pyplot as plt
N=20
y=np.zeros(N)
for n in range(N):
    if n==0:
        y[n]=1
    else:
        y[n]=1+0.5*y[n-1]
plt.figure(figsize=(8,4))
plt.stem(np.arange(N),y)
plt.xlabel("n")
plt.ylabel("Output")
plt.title("Difference Equation Solution")
plt.grid(True)
plt.show()
```



Problem 14

```
import numpy as np
import matplotlib.pyplot as plt
n=np.arange(0,20)
h1=(0.3)**n
h2=(0.8)**n
plt.figure(figsize=(8,4))
plt.stem(n,h1,
         label="Pole=0.3",
         basefmt=" ")
plt.stem(n,h2,
         linefmt='r-',
         markerfmt='ro',
         label="Pole=0.8",
         basefmt=" ")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.title("Comparison of Impulse Responses")
plt.grid(True)
plt.legend()
plt.show()
```

